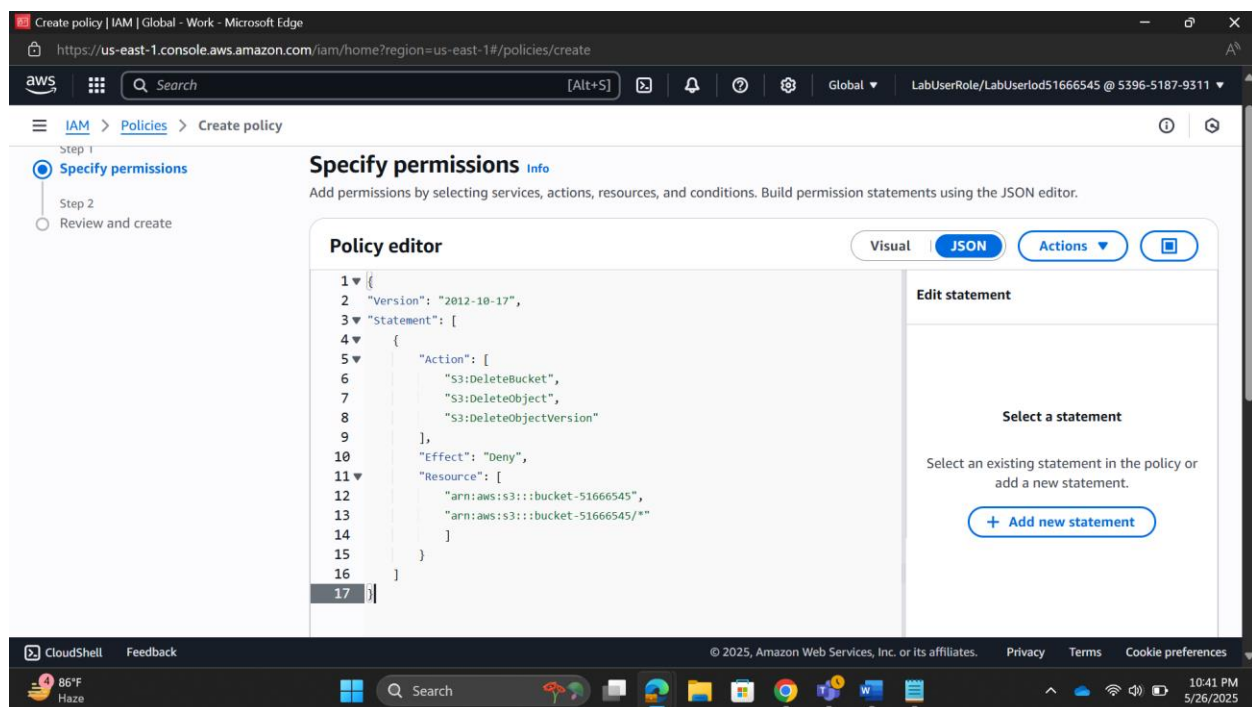


Lab 3 Implement an IAM Policy

In this challenge lab, you will create an AWS Identity and Access Management (IAM) policy to manage an Amazon Simple Storage Service (Amazon S3) bucket. First, you will create an IAM policy, and then you will create a user group. Next, you will add a user account to the group, and then you will create an S3 bucket. Finally, you will add an object to the bucket, and then you will attempt to delete both the object and the bucket. Note: Once you begin the challenge lab, you will not be able to pause, save, or exit and then return to your challenge lab. Please ensure that you have set aside enough time to complete the challenge lab before you start.

Task 1: Create a policy to prevent the deletion of an S3 bucket and its objects

IAM -> access management -> policies -> create policies -> JSON format



The screenshot shows the AWS IAM console 'Create policy' page in the 'JSON' tab. The page is titled 'Specify permissions' and includes a sidebar with steps: 'Specify permissions' (selected), 'Step 2', and 'Review and create'. The main content area shows a JSON policy document in the 'Policy editor'.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "s3:DeleteBucket",
7         "s3:DeleteObject",
8         "s3:DeleteObjectVersion"
9       ],
10      "Effect": "Deny",
11      "Resource": [
12        "arn:aws:s3::bucket-51666545",
13        "arn:aws:s3::bucket-51666545/*"
14      ]
15    }
16  ]
17 }
```

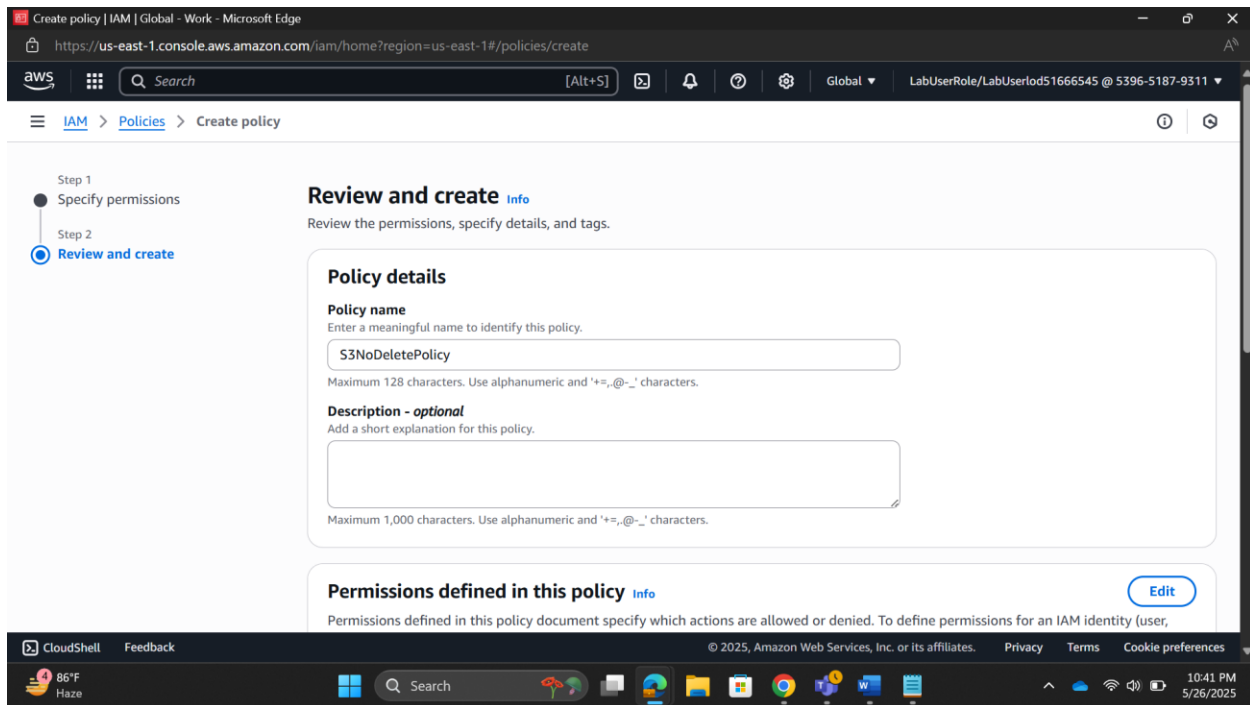
On the right side, there is an 'Edit statement' section with the text 'Select a statement' and 'Select an existing statement in the policy or add a new statement.' Below this is a button labeled '+ Add new statement'.

The bottom of the screenshot shows the Windows taskbar with the date and time as 10:41 PM on 5/26/2025.

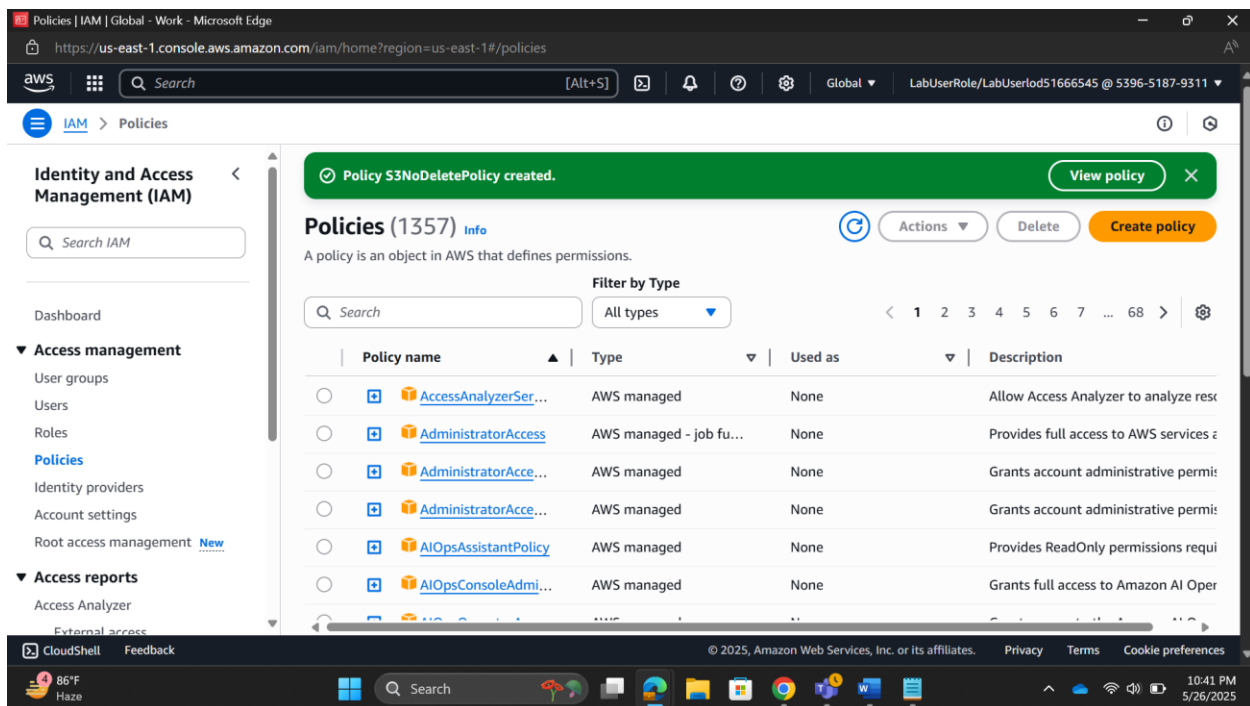
JSON Policy Code:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "S3:DeleteBucket",
        "S3:DeleteObject",
        "S3:DeleteObjectVersion"
      ],
      "Effect": "Deny",
      "Resource": [
        "arn:aws:s3:::bucket-51666545",
        "arn:aws:s3:::bucket-51666545/*"
      ]
    }
  ]
}
```

Click next



Create policy



Task 2: Create an IAM group

IAM -> Access management -> Users group -> create group

Create user group | IAM | Global - Work - Microsoft Edge

https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups/create

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management **New**

▼ Access reports

Access Analyzer

External access

Create user group

Name the group

User group name
Enter a meaningful name to identify this group.

Developers

Maximum 128 characters. Use alphanumeric and '+@_.' characters.

Add users to the group - Optional (1/1) Info

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☒	User name	Groups	Last activity	Creation time
☒	User1-51666545	0	None	5 minutes ago

Attach permissions policies - Optional (1052) Info

Create user group | IAM | Global - Work - Microsoft Edge

https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups/create

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management **New**

▼ Access reports

Access Analyzer

External access

Create user group

Add users to the group - Optional (1/1) Info

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☒	User name	Groups	Last activity	Creation time
☒	User1-51666545	0	None	5 minutes ago

Attach permissions policies - Optional (1/1052) Info

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

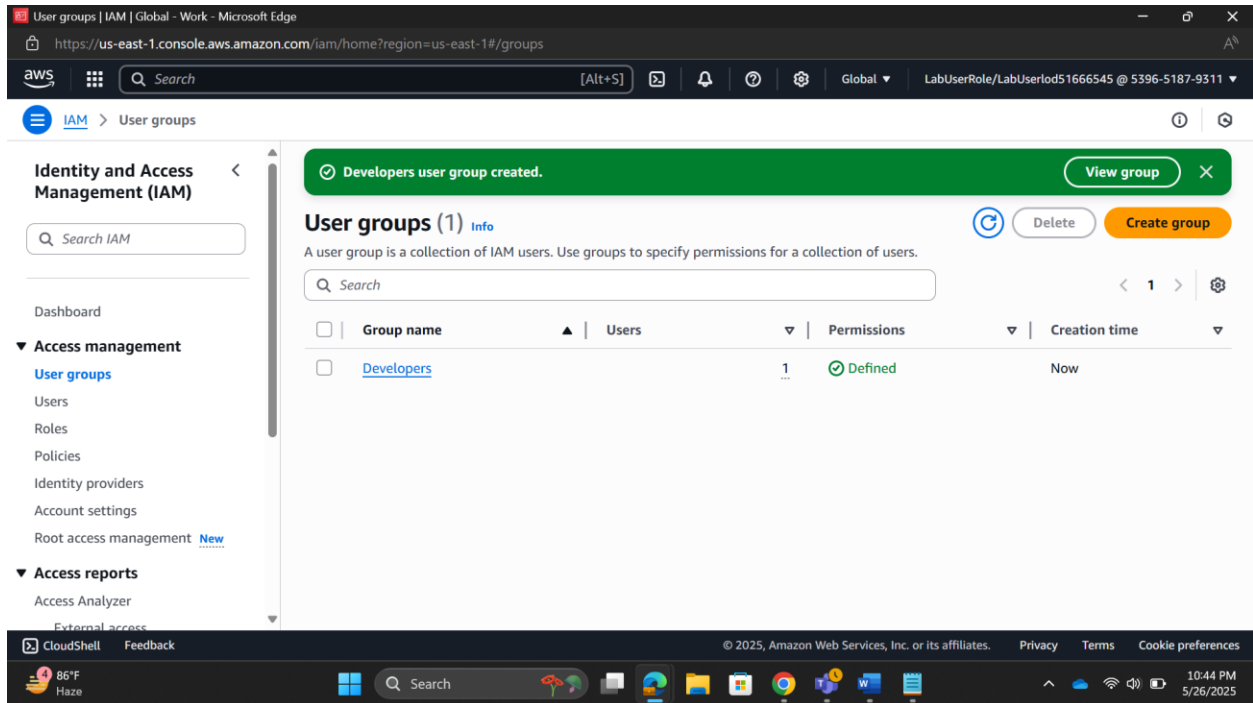
Filter by Type

Search: S3NoDeletePolicy

All types 1 match

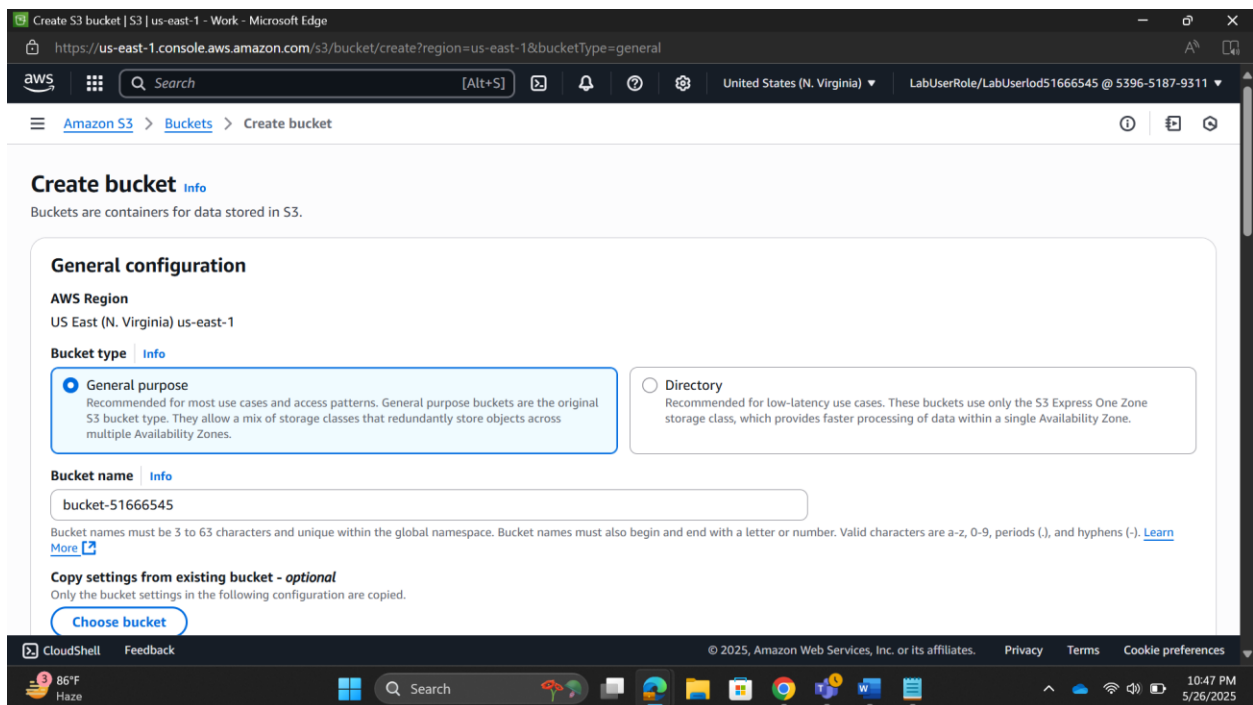
☒	Policy name	Type	Used as	Description
☒	S3NoDeletePolicy	Customer managed	None	-

Cancel Create user group



Task 3: Create an S3 bucket that contains an object

S3 -> create bucket



Click Create bucket

Upload any file to the bucket

bucket-51666545 Info

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
test1.txt	txt	May 26, 2025, 22:48:28 (UTC+05:30)	19.0 B	Standard

Task 4: Test the IAM policy

bucket-51666545 Info

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For other users to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
test1.txt	txt	May 26, 2025, 22:48:28 (UTC+05:30)	19.0 B	Standard

Account ID
5396-5187-9311

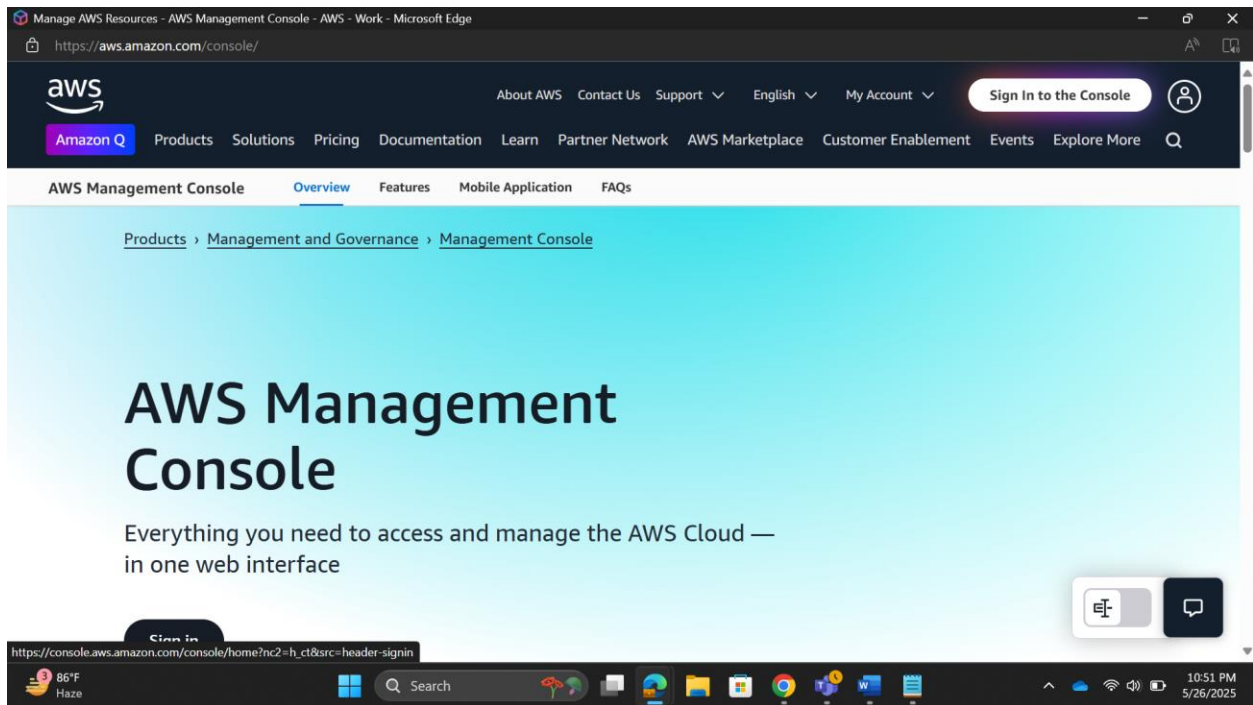
Federated user
LabUserRole/LabUserLod51666545

Account
Organization
Service Quotas
Billing and Cost Management

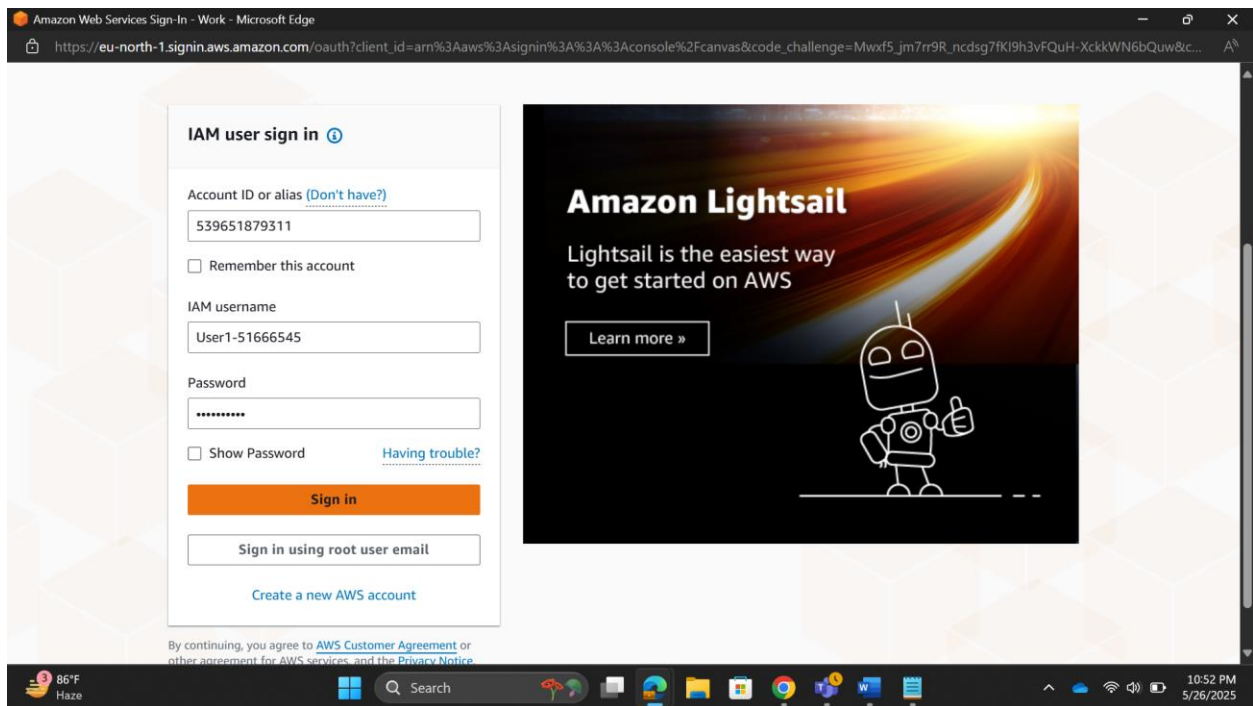
[Turn on multi-session support](#)

[Switch role](#) [Sign out](#)

Sign out from your current account



sign in to the console as IAM user sign in



Click sign in

S3 -> select the bucket created -> select the file in bucket -> delete

bucket-51666545 - S3 bucket | S3 | us-east-1 - Work - Microsoft Edge

https://us-east-1.console.aws.amazon.com/s3/buckets/bucket-51666545?region=us-east-1&tab=objects&bucketType=general

Amazon S3 > Buckets > bucket-51666545

bucket-51666545 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (1/1) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☒	Name	Type	Last modified	Size	Storage class
☒	test1.txt	txt	May 26, 2025, 22:48:28 (UTC+05:30)	19.0 B	Standard

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

86°F Haze 10:55 PM 5/26/2025

Delete objects - S3 bucket bucket-51666545 | S3 | us-east-1 - Work - Microsoft Edge

https://us-east-1.console.aws.amazon.com/s3/buckets/bucket-51666545/object/delete?region=us-east-1&showversions=false

Amazon S3 > Buckets > bucket-51666545 > Delete objects

selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.

- Deleting the specified objects can't be undone.

[Learn more](#)

Specified objects

Find objects by name

Name	Type	Last modified	Size
test1.txt	txt	May 26, 2025, 22:48:28 (UTC+05:30)	19.0 B

Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

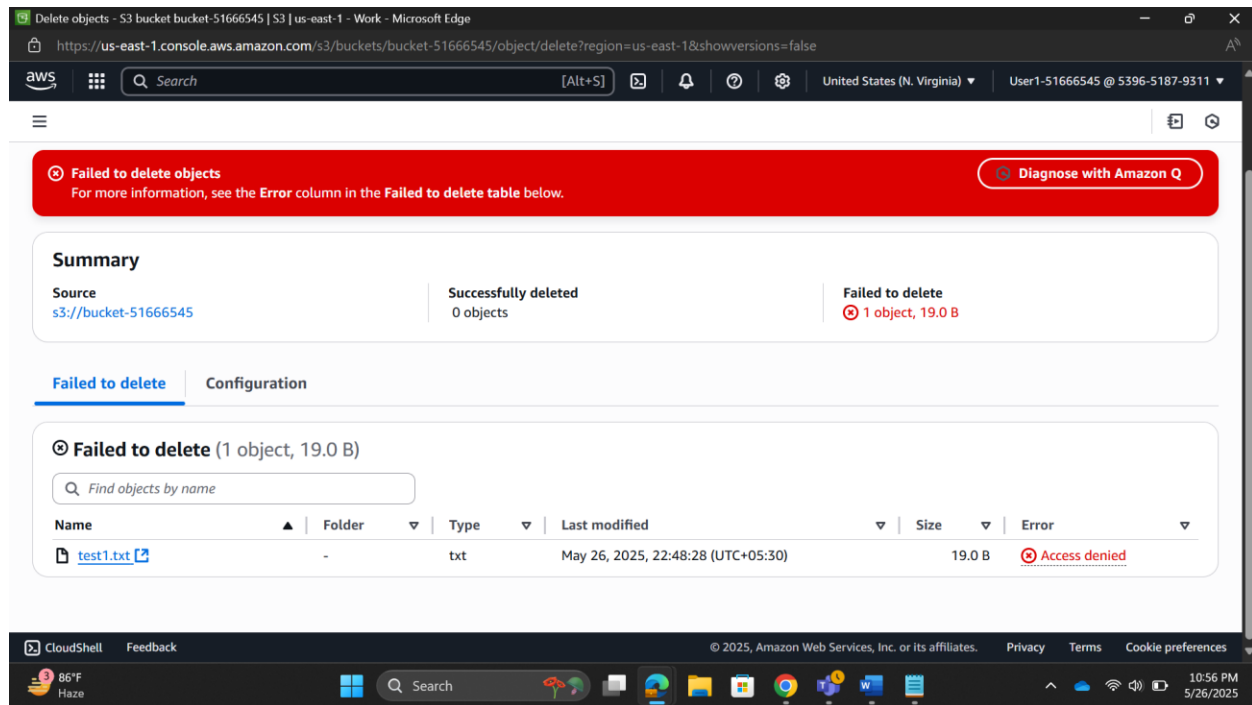
[Cancel](#) [Delete objects](#)

CloudShell Feedback

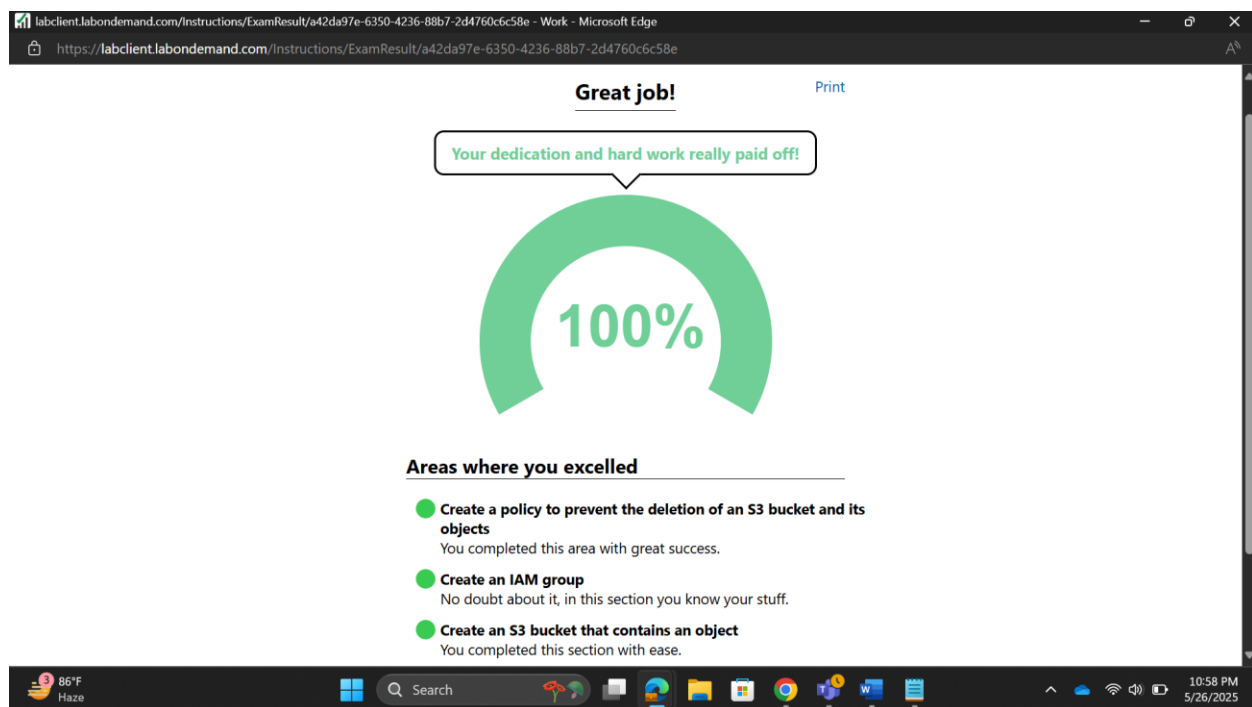
© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

86°F Haze 10:55 PM 5/26/2025

Click delete objects



Access denied because of the policy we created which is denying the deletion of file from the bucket



You are done with the lab!!!!